Gmail **Dhanshri sonawane <dhanshrisonawane0507@gmail.com>**

Quantum Graph Neural Network

Dr. Syed Muzamil basha <muzamilbasha.s@reva.edu.in> Sat, Oct 24, 2026 at 3:52 PM
To: Dhanshri sonawane <dhanshrisonawane0507@gmail.com>

Responses to Research Planning and Manuscript Revision Comments

1. Which experiment should the paper report: the Qiskit run (Figs. 5–11), the PennyLane run (noise-sweep table), or both as separate experiments?

Response: Report both experiments as separate experimental studies only if their configurations, datasets, noise models, circuit architectures, and evaluation protocols can be documented and verified. The Qiskit results (Figs. 5–11) and PennyLane noise-sweep results should not be merged or presented as though they came from the same implementation. First, establish reproducibility and consistency between the two implementations. If they use materially different configurations, report the PennyLane results as an independent validation or exploratory experiment, clearly identifying the differences. The primary results should come from the experiment with the strongest reproducibility, statistical support, and alignment with the paper's central hypothesis.

2. Should we keep the current negative-result framing, or test the finite-noise optimum hypothesis with a stronger setup?

Response: Retain the current negative-result findings as the preliminary evidence, but strengthen the experimental design to test the finite-noise optimum hypothesis before finalizing the manuscript. The existing results show that increasing noise reduces the generalization gap and output sensitivity, but also increases training and test errors. Therefore, they do not establish that noise improves generalization or test accuracy.

Conduct a more rigorous noise sweep with multiple random seeds, a fixed training/validation/test split, validation-based noise selection, confidence intervals, and suitable classical baselines. Investigate whether an intermediate noise level improves test performance consistently. If no statistically supported finite-noise optimum is observed, retain the negative-result framing and present it as evidence of noise-induced capacity suppression under the tested conditions. Do not force a positive result.

3. Which novel contribution should be selected: A–G?

Response: Prioritize **D — scale-free complexity**, followed by **A — dual-noise training**, subject to a focused literature review and feasibility assessment.

Contribution D should be developed first because it directly supports the paper's central research question: how to distinguish beneficial regularization from destructive capacity suppression in noisy quantum graph neural networks. The proposed complexity measure must be precisely defined, numerically stable, reproducible, and empirically validated against independent diagnostics such as output sensitivity, generalization gap, and test error. It should not be presented as a reliable capacity measure solely because its numerical value decreases with noise.

Contribution A can follow if the theoretical motivation and computational budget support a clear experimental comparison. Compare the proposed dual-noise approach against noiseless training, fixed-noise training, and appropriate regularization baselines. Before committing to either contribution, verify that the selected approach offers a defensible novelty beyond existing finite-noise generalization studies.

4. What compute is available?

Response: The current computational evidence indicates that the PennyLane sweep required approximately 139 minutes for 10 small runs. We should therefore avoid committing to large-scale sweeps until the runtime has been profiled and the available resources confirmed.

Initially, optimize the existing implementation and conduct a small reproducibility pilot. Consider batching, reducing unnecessary simulator overhead, and using an appropriate faster simulator or institutional HPC resources where available. Any acceleration must preserve the intended noise model and measurement semantics. The final experimental plan should specify the available CPU/GPU resources, memory, runtime budget, number of random seeds, noise levels, and model configurations. We should not claim access to a GPU cluster or assume that a GPU will accelerate the chosen simulator without benchmarking it.

5. May I edit the paper directly, or should I send a revision list for approval first?

Response: Please prepare a section-wise revision list for approval before making substantial changes to the manuscript. The list should identify the proposed revisions to the research hypothesis, theoretical contribution, experimental design, results interpretation, figures, and limitations. Clearly distinguish changes supported by existing evidence from those that depend on additional experiments. After the revision plan is approved, proceed with the manuscript edits while preserving a record of changes and ensuring that every quantitative claim is traceable to reproducible results.

Proposed overall direction: Validate the Qiskit and PennyLane results independently, strengthen the finite-noise experiment, prioritize contribution D, and retain the negative-result interpretation unless the new experiments demonstrate a statistically credible improvement in test performance. This approach provides a more rigorous foundation for the manuscript and avoids overstating the benefits of quantum noise.

Thank you and Regards,
Dr. Syed Muzamil Basha (MACM, LT-ISTE, LT-IOASD)
Professor | School of Computer Science and Engineering
Email: muzamilbasha.s@reva.edu.in | muza.basha@gmail.com | Ph: 080-46966966, | Fax: 080-46966966|Mobile:-8331977568
Industry based Sponsored Research consultant| POST DOC Fellow | University of Ha'il **|** Kingdom of Saudi Arabia

Dr. Syed Muzamil Basha – Academic Portfolio

REVA UNIVERSITY | Rukmini Knowledge Park | Kattigenahalli | Yelahanka | Bengaluru | Karnataka 560 064









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Department of Computer Science and Engineering aspires to create a pool of high-calibre technologists and researchers in the field of computer science and engineering who have potential to contribute for development of the nation and society with their expertise, skills, innovative problem-solving abilities and strong ethical values.

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MD2 Provide quality education in both theoretical and applied foundations of computer science and engineering, related inter-disciplinary areas and train students to effectively apply the knowledge to solve real-world problems.

MD3 Amplify student's potential for life-long high-quality careers and make them competitive in ever-changing and challenging global work environment.

MD4 Forge research and academic collaboration with industries and top global universities in order to provide students with greater opportunities.

MD5 Support the society by encouraging and participating in technology transfer.

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PSO-2: Solve latest problems and develop code to address the requirements of Industry through programming.

PSO-3: Use modern tools and techniques in the area of Computer Science and Engineering.

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Identify, formulate, research literature, and analyse engineering problems to arrive at substantiated conclusions using first principles of Mathematics, Natural, and Engineering Sciences.

PO3. Design/development of solutions:
Design solutions for complex engineering problems and design system components, processes to meet the specifications with consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems:
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Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

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Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

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Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings.

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Communicate effectively with the engineering community and with society at large. Be able to comprehend and write effective reports documentation. Make effective presentations and give and receive clear instructions.

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Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

From: Dhanshri sonawane <dhanshrisonawane0507@gmail.com>
Sent: 23 October 2026 14:08

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